The Dirichlet Tensor on Containers

Extracted from a machine-checked development

Abstract

We equip the category **Cont** of containers with the Dirichlet tensor

$$C \otimes D \;=\; \big(S \times T,\ (s,t) \mapsto P(s) \times Q(t)\big)$$

and unit y (one shape, one position—the same container that is the unit for sequential composition). Bifunctoriality, structural isomorphisms, naturality, and the Mac Lane coherence laws all hold by direct computation. The extension of a Dirichlet tensor has a useful normal form but is *not* a pointwise combination of the factor extensions; the honest monoidal comparison is Day convolution of $(\mathbf{Set}, \times, 1)$, with a coend isomorphism (Niu–Spivak). We also record the uniformity morphism $C \otimes D \to C \triangleleft D$ embedding the Dirichlet tensor as the non-dependent fragment of sequential composition, and prove it is not an isomorphism in general. The proofs of all stated results have been machine-checked; we record only the mathematical content.

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1 Introduction

Containers present polynomial endofunctors. Among the monoidal structures on the category **Cont** of containers, two share shapes $S \times T$ and differ only in positions:

$$C \times D \;=\; (S \times T,\ (s,t) \mapsto P(s) \oplus Q(t)) \qquad \text{(categorical product)}, \tag{1}$$

$$C \otimes D \;=\; (S \times T,\ (s,t) \mapsto P(s) \times Q(t)) \qquad \text{(Dirichlet tensor)}. \tag{2}$$

That single swap—sum of positions versus product of positions—is the heart of the chapter. On extensions it is the difference between a pointwise product of functors (by the exponential law $X^{A+B} \cong X^A \times X^B$) and a structure that does *not* split pointwise (witness: $y^3 \otimes y^3 = y^9$, while anything pointwise would give y^6).

The unit of the Dirichlet tensor is y, the one-shape one-position container—definitionally the same as the unit I of sequential composition $\triangleleft$. Two of the four monoidal structures on **Cont** thus share a unit; both restrict to the identity on that container.

What this note claims.

1. $(\mathbf{Cont}, \otimes, y)$ is a monoidal category, with all laws transport-free.
2. The extension of $C \otimes D$ has the normal form $\sum_{(s,t)} X^{P(s)\times Q(t)}$, a definitional equality of types and *not* a strict-monoidality statement.
3. The uniformity morphism $C \otimes D \to C \triangleleft D$ is canonical but not an isomorphism in general.

We work throughout in a single universe of types. Proofs are indicated by the phrase "by lean."

2 The tensor and its unit

Definition 2.1 (Container)**.** A **container** $C = (S, P)$ consists of a shape type S and a position family P. Morphisms are shape maps with contravariant position maps; we write **Cont** for the category.

Definition 2.2 (Dirichlet tensor)**.** The **Dirichlet tensor** of containers $C = (S, P)$ and $D = (T, Q)$ is the container

$$C \otimes D \;=\; \big(S \times T,\; (s,t) \mapsto P(s) \times Q(t)\big). \tag{3}$$

Positions are a *product*—this is exactly what distinguishes $\otimes$ from the categorical product, whose positions are the *coproduct* $P(s) \oplus Q(t)$.

Definition 2.3 (Unit)**.** The unit for the Dirichlet tensor is the container

$$y \;=\; (1,\; * \mapsto 1) \tag{4}$$

—one shape, one position. It is the same container as the unit I of the sequential operator $\triangleleft$; the name y is the polynomial reading as the representable y^1.

Remark 2.4. The left and right unitors for $\otimes$ are unit-collapses in both shape and position components: $y \otimes C$ has shapes $1 \times S$ and positions $1 \times P(s)$.

3 Bifunctoriality

Definition 3.1 (Action on morphisms)**.** Given morphisms $\mu : C \to C'$ and $\nu : D \to D'$, the **Dirichlet action**

$$\mu \otimes_2 \nu \;:\; C \otimes D \;\longrightarrow\; C' \otimes D'$$

acts componentwise:

$$\text{on shapes:} \quad (s,t) \;\longmapsto\; \big(\mu(s),\, \nu(t)\big), \tag{5}$$

$$\text{on positions:} \quad (p,q) \;\longmapsto\; \big(\mu_s^\sharp(p),\, \nu_t^\sharp(q)\big). \tag{6}$$

Unlike sequential composition, whose morphism action threads the outer position map through the inner shape assignment, the Dirichlet action is strictly componentwise.

Proposition 3.2 (Functor laws)**.** *1.* $\mathrm{id}_C \otimes_2 \mathrm{id}_D = \mathrm{id}_{C\otimes D}$*;*

2. $(\mu\,;\mu') \otimes_2 (\nu\,;\nu') = (\mu \otimes_2 \nu)\,;(\mu' \otimes_2 \nu')$.

Proof. By lean. □

4 Structural isomorphisms and coherence

Definition 4.1 (Associator and unitors)**.** 1. The **associator** $(C \otimes D) \otimes E \xrightarrow{\sim} C \otimes (D \otimes E)$ is product reassociation on shapes and (backward) product reassociation on positions. Product reassociation is its own mirror image, so the position component is transport-free.

2. The **left unitor** $y \otimes C \xrightarrow{\sim} C$ and **right unitor** $C \otimes y \xrightarrow{\sim} C$ are the unit-collapses in both components.

Both round trips of each isomorphism hold by definitional η for products and the unit type.

Proposition 4.2 (Naturality)**.** *The associator and both unitors of the Dirichlet tensor are natural in all arguments.*

Proof. By lean. □

Theorem 4.3 (Coherence)**.** *The pentagon and triangle identities hold for the Dirichlet tensor.*

Proof. By lean. □

Remark 4.4. Every map in sight is a product/unit shuffle. Definitional η for structures identifies the two legs of each coherence diagram on the nose.

Theorem 4.5 ($(\mathbf{Cont}, \otimes, y)$ is monoidal)**.** *The category of containers, equipped with the Dirichlet tensor and unit y, is a monoidal category.*

Proof. By lean. □

Remark 4.6. As with the coproduct and product, a formal development that admits only one canonical monoidal instance on **Cont** packages the Dirichlet structure as named data rather than a global instance. The sequential operator is preferred as the canonical instance because its comonoids are directed containers. Mathematically, four monoidal structures on one category is the point.

5 Extension: a normal form, not a strictness theorem

Theorem 5.1 (Normal form)**.** *For all containers $C = (S, P)$, $D = (T, Q)$ and every type X,*

$$\mathrm{Ext}(C \otimes D)\, X \;=\; \sum_{(s,t)\,:\,S \times T} \big(P(s) \times Q(t) \to X\big) \tag{7}$$

as a definitional equality of types.

Proof. By lean. □

Remark 5.2 (What this is not)**.** Theorem 5.1 is a useful computation rule and nothing more. It is *not* a monoidal-functor statement, and in particular it does not say that $\mathrm{Ext}(-)$ is strict monoidal for $\otimes$.

A comparison map must relate $\mathrm{Ext}(C \otimes D)$ to $\mathrm{Ext}\, C$ and $\mathrm{Ext}\, D$ through a tensor on the target category. The right-hand side above is not built from the functors $\mathrm{Ext}\, C$ and $\mathrm{Ext}\, D$ at all: it re-reads the container's presentation (S, T, P, Q). Observing that $\mathrm{Ext}(-)$ "preserves" a tensor that was itself defined by reading the presentation is circular; the equality is an unfolding of a definition.

Remark 5.3 (Honest comparison: Day convolution)**.** Compare the three comparisons with real content:

- sequential: $\mathrm{Ext}(G \triangleleft F)\, X \cong \mathrm{Ext}\, G(\mathrm{Ext}\, F\, X)$ (target tensor = composition of functors; a genuine iso);
- coproduct: $\mathrm{Ext}(C + D)\, X \cong \mathrm{Ext}\, C\, X \oplus \mathrm{Ext}\, D\, X$ (target tensor = pointwise coproduct);
- product: $\mathrm{Ext}(C \times D)\, X \cong \mathrm{Ext}\, C\, X \times \mathrm{Ext}\, D\, X$ (target tensor = pointwise product).

The correct intrinsic account of $\otimes$ is due to Niu and Spivak: $\otimes$ is the **Day convolution** on $[\mathbf{Set}, \mathbf{Set}]$ of the cartesian monoidal structure $(\mathbf{Set}, \times, 1)$. The comparison map $\mathrm{Ext}\, C \odot \mathrm{Ext}\, D \to \mathrm{Ext}(C \otimes D)$ is the co-Yoneda/density isomorphism out of a coend—an isomorphism, but emphatically not an identity. Thus $\mathrm{Ext}(-)$ is **strong**, not strict, monoidal for $\otimes$.

Remark 5.4 (Why $\otimes$ is non-pointwise)**.** Being a Day convolution is *not* what makes $\otimes$ non-pointwise. The categorical product $\times$ on **Cont** is also a Day convolution—of $(\mathbf{Set}, +, 0)$—and it *is* pointwise. The real criterion is whether the corepresentable splits the domain tensor:

- $y^{A+B} \cong y^A \times y^B$ — so Day-of-$+$ (the product $\times$) is pointwise;
- $y^{A \times B}$ admits **no** such splitting — so Day-of-$\times$ (the Dirichlet $\otimes$) is not.

Witness: $y^3 \otimes y^3 = y^9$, whereas anything pointwise would give y^6.

Taxonomy of the four monoidal structures on Cont.

structure	Day convolution of	unit	Ext pointwise?
product $\times$	$(\mathbf{Set}, +, 0)$	$1 = y^0$	yes
Dirichlet $\otimes$	$(\mathbf{Set}, \times, 1)$	$y = y^1$	no
coproduct $+$	not a Day convolution	0	yes
sequential $\triangleleft$	not a Day convolution	$I = y$	no

6 Dirichlet as the uniform fragment of sequential composition

Both $C \otimes D$ and $C \triangleleft D$ "put a D at every C-position", but

- $C \triangleleft D$ lets the inner D-shape *depend* on the outer C-position: its shapes are $\sum_{s\,:\,S}(P(s) \to T)$;
- $C \otimes D$ forces one inner shape *uniformly*: its shapes are $S \times T$.

Choosing the constant assignment $p \mapsto t$ therefore embeds $\otimes$ into $\triangleleft$. On positions the two agree definitionally:

$$\sum_{p\,:\,P(s)} Q\big((* \mapsto t)(p)\big) \;=\; P(s) \times Q(t). \tag{8}$$

Definition 6.1 (Uniformity morphism)**.** The canonical **uniformity morphism**

$$C \otimes D \;\longrightarrow\; C \triangleleft D$$

sends the shape (s, t) to $\langle s,\ * \mapsto t \rangle$ (the constant inner-shape assignment); on positions it is the identity, the fibres agreeing definitionally.

Theorem 6.2 (Uniformity is not an isomorphism)**.** *The uniformity morphism is not an isomorphism in general. Concretely, for $C = y^2$ (one shape, two positions) and $D = 2$ (two shapes, no positions), the shape map*

$$2 \;\longrightarrow\; (2 \to 2), \qquad t \;\longmapsto\; (* \mapsto t)$$

misses the identity function on the two-element set, hence is not surjective on shapes. Equivalently: $y^2 \otimes 2$ has two shapes while $y^2 \triangleleft 2$ has four; both have no positions, so their extensions at any X have two and four elements respectively.

Proof. By lean. □

Remark 6.3. Thus $\otimes \neq \triangleleft$: the uniformity morphism is a genuine comparison, not an equivalence. The Dirichlet tensor is sequential composition with the dependency switched off—the uniform (non-dependent) fragment of $\triangleleft$.

7 Discussion

What has been established. The Dirichlet tensor is a monoidal structure on **Cont**, transport-free and coherent. Its extension has a clean normal form that must not be misread as strict monoidality. The honest comparison is Day convolution. The uniformity morphism relates $\otimes$ to $\triangleleft$ without collapsing them.

Scope. The coend comparison of Niu–Spivak is not formalised here (it requires quotients beyond a pure core type theory). The taxonomy of the four structures is stated for orientation; the coproduct and product monoidal data are in a companion note, and the sequential structure in another.

Machine checking. Every definition and theorem statement above corresponds to a declaration in a machine-checked formalisation. The phrase "by lean" indicates that the proof has been verified by a proof assistant.

Acknowledgements

The mathematical content follows the theory of containers of Abbott, Altenkirch, and Ghani, and the Dirichlet/Day-convolution account of polynomial functors due to Niu and Spivak. The presentation here is a prose rendering of a formal development intended for human review of the theorem statements.

References

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